Chapter 10. Separation Theorems in the Plane

Jordan separation, invariance of domain, Jordan curve theorem, graphs in the plane, winding numbers

Contents

10.1 The Jordan separation theorem **2**

10.2 Invariance of domain **3**

10.3 The Jordan curve theorem **4**

10.4 Imbedding graphs in the plane **6**

10.5 Winding number of simple closed curves **7**

10.6 The Cauchy integral formula **8**

10.1 The Jordan separation theorem

Motivation. Every simple closed curve in the plane separates it into an inside and an outside. The statement is obvious for circles and polygons but surprisingly hard for wild curves; fundamental groups provide the proof. We begin with the easier half, that the complement is disconnected.

Throughout, open subsets of S^2 and $\mathbb{R}^2$ are locally path connected, so their components are open and coincide with their path components. A **separates** a connected space X if $X - A$ is not connected, and separates it into n components if $X - A$ has n components. An **arc** is a space homeomorphic to $[0,1]$; a **simple closed curve** is a space homeomorphic to S^1.

Lemma 10.1 (61.1)

Let $C \subset S^2$ be compact, $b \in S^2 - C$, and $h : S^2 - b \to \mathbb{R}^2$ a homeomorphism. If U is a component of $S^2 - C$ not containing b, then $h(U)$ is a bounded component of $\mathbb{R}^2 - h(C)$; if U contains b, then $h(U - b)$ is the unbounded component. In particular $S^2 - C$ and $\mathbb{R}^2 - h(C)$ have the same number of components.

Proof. First, $U - b$ is connected for every component U. Suppose $b \in U$ and A, B separate $U - b$. Choose a neighbourhood W of b disjoint from C and homeomorphic to an open disc; $W \subset U$, and the connected set $W - b$ lies in A, say. Then b is not a limit point of B, and $A \cup \{b\}$, B separate U, a contradiction. Hence the sets $h(U_\alpha - b)$, for the components U_α of $S^2 - C$, are disjoint connected open sets with union $\mathbb{R}^2 - h(C)$; they are its components. Extend h to a homeomorphism H of S^2 with the one-point compactification $\mathbb{R}^2 \cup \{\infty\}$ by $H(b) = \infty$. For the component $U \ni b$, $H(U)$ is a neighbourhood of ∞, so $h(U - b)$ contains the complement of a compact set: it is unbounded, and all the other components lie in a compact set, hence are bounded. ∎

Lemma 10.2 (61.2, nulhomotopy lemma)

Let $a, b \in S^2$, A compact and $f : A \to S^2 - a - b$ continuous. If a and b lie in the same component of $S^2 - f(A)$, then f is nulhomotopic.

Proof. Identify S^2 with $\mathbb{R}^2 \cup \{\infty\}$, a with 0 and b with ∞. We must show: if $g : A \to \mathbb{R}^2 - 0$ and 0 lies in the unbounded component of $\mathbb{R}^2 - g(A)$, then g is nulhomotopic. Choose a ball B about 0 containing $g(A)$ and a point p outside B; 0 and p lie in the unbounded component, so there is a path α in $\mathbb{R}^2 - g(A)$ from 0 to p. Then $G(x,t) = g(x) - \alpha(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from g to $k(x) = g(x) - p$, and $H(x,t) = tg(x) - p$ is a homotopy in $\mathbb{R}^2 - 0$ from the constant $-p$ to k (since $tg(x) \in B$ and $p \notin B$). ∎

Theorem 10.3 (61.3, Jordan separation theorem)

A simple closed curve C in S^2 separates S^2.

Proof. Suppose $S^2 - C$ is connected, hence path connected. Write $C = A_1 \cup A_2$ with arcs meeting only in their end points a, b, and let $X = S^2 - a - b$, $U = S^2 - A_1$, $V = S^2 - A_2$. Then $X = U \cup V$ and $U \cap V = S^2 - C$ is path connected. We show $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ is trivial; similarly j_*; then $\pi_1(X, x_0)$ is trivial by Theorem 9.38, whereas $X \cong \mathbb{R}^2 - 0$. Let f be a loop in U at x_0; it induces $h : S^1 \to U$ with $h \circ p_0 = f$. The set $h(S^1)$ misses the connected set $A_1 \ni a, b$, so a and b lie in one component of $S^2 - h(S^1)$, and $i \circ h$ is nulhomotopic (Lemma 10.2). By Lemma 9.20, $(i \circ h)_*$ is trivial, and $i_*[f] = (i \circ h)_*[p_0]$. ∎

Theorem 10.4 (61.4, general separation theorem)

If A_1, A_2 are closed connected subsets of S^2 with $A_1 \cap A_2 = \{a, b\}$, then $C = A_1 \cup A_2$ separates S^2.

Proof. $C \neq S^2$, since $S^2 - a - b$ is connected while $C - a - b$ is separated by the sets $A_i - a - b$. The rest of the proof of Theorem 10.3 used only that A_1, A_2 are closed and connected and meet in $\{a, b\}$. ■

10.2 Invariance of domain

Motivation. If $U \subseteq \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open. This is the two-dimensional case of Brouwer's theorem and is needed in the Jordan curve theorem and in the theory of manifolds.

Lemma 10.5 (62.1, homotopy extension lemma)

Let $X \times I$ be normal, $A \subset X$ closed, and $f : A \to Y$ continuous with Y open in $\mathbb{R}^n$. If f is nulhomotopic, it extends to a continuous nulhomotopic $g : X \to Y$.

Proof. Let $F : A \times I \to Y$ be a homotopy from f to the constant y_0, and set $F(x, 1) = y_0$ for $x \in X$. This is continuous on the closed set $(A \times I) \cup (X \times 1)$, so by Tietze (coordinatewise) it extends to $G : X \times I \to \mathbb{R}^n$. The open set $U = G^{-1}(Y)$ contains $(A \times I) \cup (X \times 1)$; by the tube lemma each $a \in A$ has a neighbourhood W_a with $W_a \times I \subset U$, and $W = \bigcup W_a$ is open with $A \subset W$, $W \times I \subset U$. $X \cong X \times 0$ is normal, so Urysohn gives $\phi : X \to [0, 1]$ with $\phi = 0$ on A, $\phi = 1$ off W. Then $(x, \phi(x)) \in U$ always, and $g(x) = G(x, \phi(x))$ maps X into Y and extends f. The homotopy $H(x, t) = G(x, (1 - t)\phi(x) + t)$ stays in Y for the same reason and joins g to the constant y_0. ■

Lemma 10.6 (62.2, Borsuk lemma)

Let $a, b \in S^2$, A compact, and $f : A \to S^2 - a - b$ continuous and injective. If f is nulhomotopic, then a and b lie in the same component of $S^2 - f(A)$.

Proof. $f(A)$ is homeomorphic to A, so we may take f to be an inclusion; identify a with 0 and b with ∞. We show: if $A \subset \mathbb{R}^2 - 0$ is compact and the inclusion $j : A \to \mathbb{R}^2 - 0$ is nulhomotopic, then 0 lies in the unbounded component of $\mathbb{R}^2 - A$. Suppose the component C containing 0 is bounded, and let D be the union of the other components; $\mathbb{R}^2 - A = C \cup D$, both open. $C \cup A = \mathbb{R}^2 - D$ is closed and bounded, hence compact, and Lemma 10.5 extends j to $k : C \cup A \to \mathbb{R}^2 - 0$. Define $h = k$ on $C \cup A$ and $h(x) = x$ on $D \cup A$; by the pasting lemma $h : \mathbb{R}^2 \to \mathbb{R}^2 - 0$ is continuous. Choose a closed ball B of radius M whose interior contains $C \cup A$. Then $h|B$ is the identity on $\operatorname{Bd} B$, and $x \mapsto Mh(x)/\|h(x)\|$ is a retraction of B onto $\operatorname{Bd} B$, contradicting Theorem 9.19. ■

Theorem 10.7 (62.3, invariance of domain)

If $U \subset \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open and $f^{-1} : f(U) \to U$ is continuous.

Proof. Regard f as a map into $S^2 \supset \mathbb{R}^2$; it suffices that f is an open map into S^2.

Step 1. for a closed ball $B \subset U$, $f(B)$ does not separate S^2. If $a, b \notin f(B)$, then $f|B : B \to S^2 - a - b$ is nulhomotopic (as B is contractible) and injective, so a, b lie in one component of $S^2 - f(B)$ (Lemma 10.6).

Step 2. $f(\operatorname{Int} B)$ is open. $C = f(\operatorname{Bd} B)$ is a simple closed curve, so it separates S^2 (Theorem 10.3). Let V be the component of $S^2 - C$ containing the connected set $f(\operatorname{Int} B)$ and $W \neq \emptyset$ the union of the others. If $a \in V - f(\operatorname{Int} B)$, then $a \notin f(B)$; pick $b \in W$. By Step 1, a, b lie in a connected subset of $S^2 - f(B) \subset S^2 - C$, so in the same component of $S^2 - C$, a contradiction. Hence $f(\operatorname{Int} B) = V$ is open.

Step 3. every open subset of U is a union of interiors of closed balls in U, so f is an open map. ■

The theorem holds in $\mathbb{R}^n$ for all n, but the proof requires higher homotopy or homology groups.

10.3 The Jordan curve theorem

Motivation. We complete the proof: the complement of a simple closed curve has exactly two components, and the curve is the boundary of each.

Theorem 10.8 (63.1)

Let $X = U \cup V$ with U, V open and $U \cap V = A \cup B$, A, B disjoint and open. Let α be a path in U from $a \in A$ to $b \in B$, β a path in V from b to a, and $f = \alpha * \beta$.
(a) $[f]$ generates an infinite cyclic subgroup of $\pi_1(X, a)$.
(b) If $\pi_1(X, a)$ is infinite cyclic, it is generated by $[f]$.
(c) If γ is a path in U from a to $a' \in A$ and δ a path in V from a' to a, and $g = \gamma * \delta$, then the subgroups generated by $[f]$ and $[g]$ meet only in the identity.

Proof.

Step 1 (a covering space). Let Y be the disjoint union of copies $U \times 2n$ and $V \times (2n+1)$, $n \in \mathbb{Z}$, inside $X \times \mathbb{Z}$, and let E be the quotient of Y obtained by identifying $x \times 2n$ with $x \times (2n+1)$ for $x \in B$ and $x \times 2n$ with $x \times (2n-1)$ for $x \in A$. Let $\pi : Y \to E$ be the quotient map; $x \times m \mapsto x$ induces a continuous surjection $p : E \to X$. π is open: for W open in U, $\pi^{-1}\pi(W \times 2n) = W \times 2n \cup (W \cap B) \times (2n+1) \cup (W \cap A) \times (2n-1)$, which is open; similarly for $V \times (2n+1)$. Hence $p^{-1}(U)$ is the disjoint union of the open sets $\pi(U \times 2n)$, and $\pi | U \times 2n$ is a continuous open bijection onto its image, so p maps each $\pi(U \times 2n)$ homeomorphically onto U; likewise for V. So p is a covering map.

Step 2. Let $e_n = \pi(a \times 2n)$; these are the distinct points of $p^{-1}(a)$. The paths $\tilde{\alpha}_n(s) = \pi(\alpha(s) \times 2n)$ and $\tilde{\beta}_n(s) = \pi(\beta(s) \times (2n+1))$ lift α, β; $\tilde{\alpha}_n$ ends at $\pi(b \times 2n) = \pi(b \times (2n+1))$, where $\tilde{\beta}_n$ begins, and $\tilde{\beta}_n$ ends at $\pi(a \times (2n+1)) = e_{n+1}$. So $\tilde{f}_n = \tilde{\alpha}_n * \tilde{\beta}_n$ lifts f from e_n to e_{n+1}.

Step 3 (a). For $m > 0$, $\tilde{f}_0 * \cdots * \tilde{f}_{m-1}$ lifts f^m from e_0 to $e_m \neq e_0$, so $[f]^m \neq 1$.

Step 4 (b). The lifting correspondence ϕ sends $[f]^m$ to e_m for all $m \in \mathbb{Z}$ (similarly for negative m), so it is surjective. By Theorem 9.17, ϕ induces a bijection $\pi_1(X, a)/H \to p^{-1}(a)$, $H = p_*\pi_1(E, e_0)$; the fibre is infinite while a quotient of $\mathbb{Z}$ by a nontrivial subgroup is finite, so H is trivial and ϕ is bijective. Since ϕ maps the subgroup generated by $[f]$ onto the fibre, this subgroup is all of $\pi_1(X, a)$.

Step 5 (c). $\tilde{\gamma}(s) = \pi(\gamma(s) \times 0)$ and $\tilde{\delta}(s) = \pi(\delta(s) \times (-1))$ lift γ, δ, and $\pi(a' \times 0) = \pi(a' \times (-1))$ as $a' \in A$; $\tilde{\delta}$ ends at $\pi(a \times (-1)) = e_0$. So g lifts to a loop at e_0, and so does every power of g, while f^m $(m \neq 0)$ lifts to a path ending at $e_m \neq e_0$. Hence $[f]^m \neq [g]^k$ unless $m = 0$, which forces $[g]^k = 1$. ■

Theorem 10.9 (63.2, nonseparation theorem)

An arc D in S^2 does not separate S^2.

Proof. D is contractible, so for $a, b \notin D$ the inclusion $D \to S^2 - a - b$ is nulhomotopic; by Lemma 10.6, a and b lie in the same component of $S^2 - D$. ∎

A second proof gives the following generalization.

Theorem 10.10 (63.3, general nonseparation theorem)

Let D_1, D_2 be closed subsets of S^2 with $S^2 - (D_1 \cap D_2)$ simply connected. If neither D_1 nor D_2 separates S^2, neither does $D_1 \cup D_2$.

Proof. Let $a, b \notin D_1 \cup D_2$ and suppose they cannot be joined by a path in $S^2 - (D_1 \cup D_2)$. Let $X = S^2 - (D_1 \cap D_2)$, $U = S^2 - D_1$, $V = S^2 - D_2$; then $X = U \cup V$ and $U \cap V = S^2 - (D_1 \cup D_2)$. Let A be the path component of $U \cap V$ containing a and B the union of the others, both open. Since D_1, D_2 do not separate, there are paths in U from a to b and in V from b to a. Theorem 10.8(a) shows $\pi_1(X, a)$ is nontrivial, a contradiction. ∎

(For an arc $D = h([0,1])$, this yields the second proof of Theorem 10.9: if a, b cannot be joined in $S^2 - D$, they cannot be joined in $S^2 - h(I_1)$ for one of the halves I_1 of $[0,1]$, since S^2−point is simply connected; bisecting repeatedly gives nested I_n shrinking to a point x, but a path from a to b in $S^2 - h(x)$ has compact image missing $h(I_n)$ for large n.)

Theorem 10.11 (63.4, Jordan curve theorem)

A simple closed curve C in S^2 separates S^2 into precisely two components W_1, W_2, and $C = \overline{W_i} - W_i$ for $i = 1, 2$.

Proof.

Step 1. write $C = C_1 \cup C_2$ with arcs meeting in $\{p, q\}$; let $X = S^2 - p - q$, $U = S^2 - C_1$, $V = S^2 - C_2$, so $U \cap V = S^2 - C$, which has at least two components (Theorem 10.3). Suppose it has more: let A_1, A_2 be two components and B the union of the others, all open; pick $a \in A_1$, $a' \in A_2$, $b \in B$. The arcs C_i do not separate S^2 (Theorem 10.9), so there are paths α, γ in U from a to b and from a to a', and β, δ in V from b to a and from a' to a. With $f = \alpha * \beta$ and $g = \gamma * \delta$, Theorem 10.8(a) applied to the decomposition $U \cap V = (A_1 \cup A_2) \cup B$ shows $[f] \neq 1$ has infinite order, and applied to $A_1 \cup (A_2 \cup B)$ shows the same for $[g]$. Since $\pi_1(X, a) \cong \mathbb{Z}$, $[f]^m = [g]^k$ for some nonzero m, k, contradicting Theorem 10.8(c) for the decomposition $(A_1 \cup A_2) \cup B$, in which $a, a' \in A_1 \cup A_2$.

Step 2. W_1, W_2 are open, so neither contains a limit point of the other, and $\overline{W_i} - W_i \subset C$. Conversely let $x \in C$ and U a neighbourhood of x. Write $C = C_1 \cup C_2$ with arcs meeting in their end points and $C_1 \subset U$. For $a \in W_1$, $b \in W_2$, take a path α from a to b in $S^2 - C_2$ (Theorem 10.9). Its connected image meets the disjoint open sets W_1 and $S^2 - \overline{W_1} \supset W_2$, so it is not contained in their union and contains a point y of $\overline{W_1} - W_1 \subset C$; as it misses C_2, $y \in C_1 \subset U$. So $x \in \overline{W_1} - W_1$, and likewise for W_2. ∎

Theorem 10.12 (63.5)

If C_1, C_2 are closed connected subsets of S^2 meeting in exactly two points, and neither separates S^2, then $C_1 \cup C_2$ separates S^2 into precisely two components.

Proof. At least two components by Theorem 10.4; at most two by Step 1 above, which used only these properties. ∎

The union of two arcs meeting in two points that are not both end points separates S^2 into two components, but the union need not be their common boundary.

10.4 Imbedding graphs in the plane

Motivation. Can three houses be connected to three utilities without crossing lines? The Jordan curve theorem shows that the graph $K_{3,3}$ cannot be embedded in the plane.

A **linear graph** is a Hausdorff space that is a finite union of arcs (**edges**), any two meeting in at most a common end point (a **vertex**). The **complete graph on n vertices** has an edge joining each pair of vertices; the **utilities graph** has vertices g, w, e, h_1, h_2, h_3 with an edge joining each of g, w, e to each h_i. A **theta space** is a Hausdorff space that is the union of three arcs A, B, C, each pair meeting precisely in their common end points.

Lemma 10.13 (64.1)

A theta space $X = A \cup B \cup C$ in S^2 separates S^2 into three components, with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The component with boundary $A \cup B$ is a component of $S^2 - (A \cup B)$.

Proof. Let a, b be the common end points. By the Jordan curve theorem, $A \cup B$ separates S^2 into U and U' with boundary $A \cup B$. The connected set $C - a - b$ lies in one, say U'. The connected sets $\overline{U} = U \cup A \cup B$ and C meet only in $\{a, b\}$, and neither separates S^2 (C is an arc, and $S^2 - \overline{U} = U'$ is connected). By Theorem 10.12, $\overline{U} \cup C$ separates S^2 into two components V, W. So $S^2 - X = U \cup V \cup W$ is a union of three disjoint open connected sets, which are its components; U has boundary $A \cup B$ and is a component of $S^2 - (A \cup B)$. By symmetry the other two components have boundaries $B \cup C$ and $A \cup C$. ■

Theorem 10.14 (64.2)

The utilities graph cannot be imbedded in the plane.

Proof. Suppose it lies in S^2. The arcs $A = gh_1 \cup h_1 w$, $B = gh_2 \cup h_2 w$, $C = gh_3 \cup h_3 w$ form a theta space, which separates S^2 into U, V, W with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The vertex e lies in one of them, and the connected sets $eh_i - h_i$ miss $A \cup B \cup C$, so lie in that component; then h_1, h_2, h_3 lie in its boundary. But $A \cup B$ misses h_3, $B \cup C$ misses h_1, $A \cup C$ misses h_2. ■

Lemma 10.15 (64.3)

A complete graph X on four vertices $a_1, \ldots, a_4$ in S^2 separates S^2 into four components, whose boundaries are the subgraphs X_i consisting of the edges not having a_i as a vertex.

Proof. Let Y be the union of all edges except a_2a_4; it is the theta space with arcs $A = a_1a_2 \cup a_2a_3$, $B = a_1a_3$, $C = a_1a_4 \cup a_4a_3$. It separates S^2 into U, V, W with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The connected set $a_2a_4 - a_2 - a_4$ lies in one of them, whose boundary contains a_2 and a_4: not U ($a_4 \notin A \cup B$), not V ($a_2 \notin B \cup C$), so W. Now $\overline{U} \cup \overline{V}$ is connected (both contain B), and does not separate S^2, its complement being W; the arc a_2a_4 does not separate S^2; and they meet only in a_2, a_4. By Theorem 10.12 their union separates S^2 into W_1, W_2. Thus $S^2 - X = U \cup V \cup W_1 \cup W_2$, and these open connected sets are its components. U has boundary $A \cup B = X_4$; by symmetry the others have boundaries X_1, X_2, X_3. ■

Theorem 10.16 (64.4)

The complete graph on five vertices cannot be imbedded in the plane.

Proof. Suppose $G \subset S^2$ has vertices $a_1, \ldots, a_5$, and let X be the complete subgraph on $a_1, \ldots, a_4$. The vertex a_5 lies in one of the four components of $S^2 - X$, and so does the connected set formed

by the edges at a_5 minus $a_1, \dots, a_4$. Hence $a_1, \dots, a_4$ all lie in the boundary of that component, which is some X_i; but X_i does not contain a_i. ■

By a theorem of Kuratowski, a graph imbeds in the plane iff it contains no subgraph homeomorphic to the utilities graph or the complete graph on five vertices.

10.5 Winding number of simple closed curves

Motivation. How many times does a curve wind around a point? The winding number, defined through the fundamental group of the punctured plane, is ± 1 inside a simple closed curve and 0 outside.

Lemma 10.17 (65.1)

Let $G \subset S^2$ be a complete graph on $a_1, \dots, a_4$, C the simple closed curve $a_1a_2a_3a_4a_1$, and p, q interior points of the edges a_1a_3 and a_2a_4. Then (a) p and q lie in different components of $S^2 - C$;
(b) the inclusion $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

Proof. (a) The theta space $C \cup a_1a_3$ (arcs $a_1a_2a_3$, a_1a_3, $a_1a_4a_3$) separates S^2 into three components; one of them, W, has boundary C, and it is the only one whose boundary contains both a_2 and a_4. So the connected set $a_2a_4 - a_2 - a_4$ lies in W and $q \in W$. By Lemma 10.13, W is a component of $S^2 - C$, and $p \notin W$ since p lies on the theta space. So p, q lie in different components of $S^2 - C$.

(b) Let $X = S^2 - p - q$. Choose x interior to a_1a_2 and y interior to a_3a_4, and let α be the path in C from x through a_2, a_3 to y and β the path in C from y through a_4, a_1 to x. Let D_1 be the arc of the simple closed curve $D = a_1a_3a_2a_4a_1$ from p through a_1, a_4 to q, and D_2 the arc from p through a_3, a_2 to q; put $U = S^2 - D_1$, $V = S^2 - D_2$. Then $X = U \cup V$, $U \cap V = S^2 - D$ has two components (Jordan curve theorem), and x, y lie in different ones by (a) applied to G with the curve D (whose "diagonals" are a_1a_2 and a_3a_4). Also $\alpha \subset U$ and $\beta \subset V$. As $\pi_1(X, x) \cong \mathbb{Z}$, Theorem 10.8(b) shows that $[\alpha * \beta]$ generates it. Since $\alpha * \beta$ is a loop in C, $j_* : \pi_1(C, x) \cong \mathbb{Z} \to \pi_1(X, x) \cong \mathbb{Z}$ is surjective, hence an isomorphism. ■

Theorem 10.18 (65.2)

If C is a simple closed curve in S^2 and p, q lie in different components of $S^2 - C$, then $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

Proof.

Step 1. if A is an arc from a to b and B an arc from b to c (a, b, c distinct), then $A \cup B$ contains an arc from a to c: with homeomorphisms $f : I \to A$, $g : I \to B$ from a to b and b to c, let t_0 be the least t with $f(t) \in B$ and $g(t_1) = f(t_0)$; then $f([0, t_0]) \cup g([t_1, 1])$ is such an arc.

Step 2. any two points of a connected open $U \subset \mathbb{R}^2$ are the end points of an arc in U: the relation "equal, or joined by an arc in U" is an equivalence relation by Step 1, with open classes (segments in a disc), hence one class.

Step 3. we build a complete graph on four vertices containing C as $a_1a_2a_3a_4a_1$. Work in $\mathbb{R}^2$ with 0 in the bounded component of $\mathbb{R}^2 - C$. Let a_1 be the largest point of C on the negative x-axis and a_3 the smallest on the positive x-axis; the open segment between them misses C and contains 0, so lies in the bounded component. Write $C = C_1 \cup C_2$, arcs from a_1 to a_3. Take a in the unbounded component. As C_1, C_2 do not separate, Step 2 gives injective paths α in $\mathbb{R}^2 - C_1$ and β in $\mathbb{R}^2 - C_2$ from a to 0. Let t_0 be least with $\alpha(t_0) \in C$ and $a_2 = \alpha(t_0)$, an interior point of C_2; similarly

$a_4 = \beta(t_1)$ interior to C_1. The arcs $\alpha([0,t_0])$ and $\beta([0,t_1])$ meet C only at their end points and otherwise lie in the unbounded component; by Step 1 their union contains an arc from a_2 to a_4 meeting $C \cup a_1a_3$ only in a_2, a_4. This arc, the segment a_1a_3 and C form the required graph. By Lemma 10.17 the theorem holds for some pair p, q in different components.

Step 4. it remains to move p (and, symmetrically, q) within its component. In $\mathbb{R}^2$-coordinates with $q = \infty$: let D be a simple closed curve, $0, p$ in the same component of $\mathbb{R}^2 - D$, and suppose $j : D \to \mathbb{R}^2 - 0$ induces an isomorphism. With $f(x) = x - p$, a homeomorphism $\mathbb{R}^2 - p \to \mathbb{R}^2 - 0$, and γ a path in $\mathbb{R}^2 - D$ from 0 to p, $F(x,t) = x - \gamma(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from j to $f \circ k$, $k : D \to \mathbb{R}^2 - p$ the inclusion. By Corollary 9.35, $(f \circ k)_*$ and hence k_* are isomorphisms. ■

10.6 The Cauchy integral formula

Motivation. Complex analysis uses the winding number implicitly. Here the topological facts proved above are used to justify the Cauchy integral formula for simple closed curves.

For a loop f in $\mathbb{R}^2$ and $a \notin f(I)$, let $g(s) = (f(s) - a)/\|f(s) - a\|$, a loop in S^1, and $\tilde{g}$ a lifting to $\mathbb{R}$. The integer $n(f,a) = \tilde{g}(1) - \tilde{g}(0)$ is the **winding number** of f about a; it does not depend on the lifting, since any two differ by an integer constant. A **free homotopy** of loops is a map $F : I \times I \to X$ with $F(0,t) = F(1,t)$ for all t.

Lemma 10.19 (66.1)

Let f be a loop in $\mathbb{R}^2 - a$.
(a) $n(\bar{f}, a) = -n(f,a)$.
(b) If f is freely homotopic to f' through loops in $\mathbb{R}^2 - a$, then $n(f,a) = n(f',a)$.
(c) If a, b lie in the same component of $\mathbb{R}^2 - f(I)$, then $n(f,a) = n(f,b)$.

Proof. (a) Replacing s by $1 - s$ reverses the sign of $\tilde{g}(1) - \tilde{g}(0)$.

(b) For a free homotopy F, lift $G(s,t) = (F(s,t) - a)/\|F(s,t) - a\|$ to $\tilde{G} : I \times I \to \mathbb{R}$ (Lemma 9.13); $\tilde{G}(1,t) - \tilde{G}(0,t)$ is a continuous integer-valued function of t, hence constant.

(c) $n(f,a) = n(f - a, 0)$, and for a path γ in $\mathbb{R}^2 - f(I)$ from a to b, $f(s) - \gamma(t)$ is a free homotopy in $\mathbb{R}^2 - 0$ from $f - a$ to $f - b$. ■

A loop f is **simple** if $f(s) = f(s')$ only when $s = s'$ or $\{s, s'\} = \{0, 1\}$; its image is a simple closed curve.

Theorem 10.20 (66.2)

Let f be a simple loop in $\mathbb{R}^2$. If a is in the unbounded component of $\mathbb{R}^2 - f(I)$, $n(f,a) = 0$; if a is in the bounded component, $n(f,a) = \pm 1$.

Proof. We may take $a = 0$, and by rotating (a free homotopy) assume the base point x_0 lies on the positive x-axis. Let $C = f(I)$ and $h : S^1 \to C$ the homeomorphism with $h \circ p_0 = f$; $h_*[p_0]$ generates $\pi_1(C, x_0)$. If 0 is in the bounded component, $[f] = j_*h_*[p_0]$ generates $\pi_1(\mathbb{R}^2 - 0, x_0)$ by Theorem 10.18 (with ∞ in the other component). If 0 is in the unbounded component, $j \circ h$ is nulhomotopic (Lemma 10.2) and $[f]$ is trivial. The retraction $x \mapsto x/\|x\|$ induces an isomorphism onto $\pi_1(S^1, b_0)$, so $g = f/\|f\|$ represents a generator or the identity, i.e. (proof of Theorem 9.16) its lifting from 0 ends at ± 1 or at 0. ■

A simple loop is **counterclockwise** if $n(f,a) = +1$ for a in the bounded component and **clockwise** if it is -1.

Lemma 10.21 (66.3)

If f is a piecewise-differentiable loop in $\mathbb{C}$ and $a \notin f(I)$, then $n(f,a) = \dfrac{1}{2\pi i}\displaystyle\int_f \frac{dz}{z-a}$.

Proof. Let $r(s) = |f(s)-a|$, $g = (f-a)/r$, $\tilde{g}$ a lifting and $\theta = 2\pi\tilde{g}$, so $f(s)-a = r(s)e^{i\theta(s)}$ with r, θ piecewise differentiable (θ is locally a branch of the argument). Then

$$\int_f \frac{dz}{z-a} = \int_0^1 \frac{f'(s)}{f(s)-a}\,ds = \int_0^1 \Big(\frac{r'(s)}{r(s)} + i\theta'(s)\Big)ds = [\log r]_0^1 + i[\theta]_0^1 = 2\pi i\, n(f,a).$$

■

Theorem 10.22 (66.4, Cauchy integral formula, classical version)

Let C be a simple closed piecewise-differentiable curve in $\mathbb{C}$, B the bounded component of $\mathbb{R}^2 - C$, and F analytic on an open set $\Omega \supset B \cup C$. For $a \in B$,

$$F(a) = \pm\frac{1}{2\pi i}\int_C \frac{F(z)}{z-a}\,dz,$$

with $+$ if C is oriented counterclockwise and $-$ otherwise.

Proof. The general version (Ahlfors): if F is analytic on a region Ω, f a piecewise-differentiable loop in Ω with $n(f,b) = 0$ for every $b \notin \Omega$, and $a \in \Omega - f(I)$, then $n(f,a)F(a) = \frac{1}{2\pi i}\int_f \frac{F(z)}{z-a}dz$. For a parametrization f of C, every $b \notin \Omega$ is in the unbounded component, so $n(f,b) = 0$, and $n(f,a) = \pm 1$ for $a \in B$ by Theorem 10.20. ■